



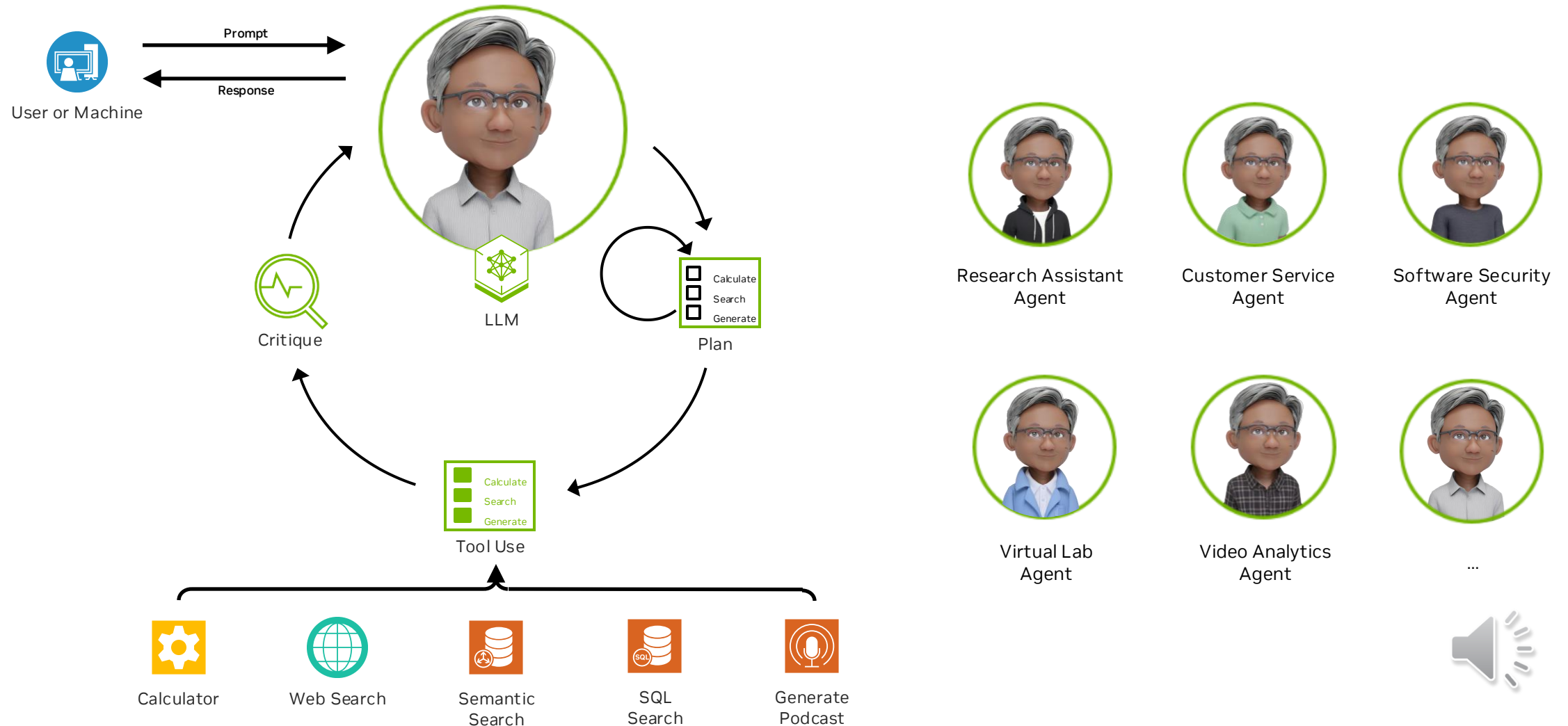
Supercharge Agentic AI: Build Scalable Data Flywheels

Chintan Patel | Shashank Verma. NVIDIA



AI Agents: The New Digital Employees

They help you get more done but require effective on-boarding



Operating AI Agents at Scale Is Challenging

Continuous Updates Are Essential for Improving Accuracy and Performance



Data Ages Rapidly

Training data, user feedback, knowledge bases update faster than training cycles



Evolving Customer Needs

Changing user behavior and preferences reduce model accuracy



Growing Inference Costs

Increased usage drives costs.
Accuracy & efficiency often at conflict



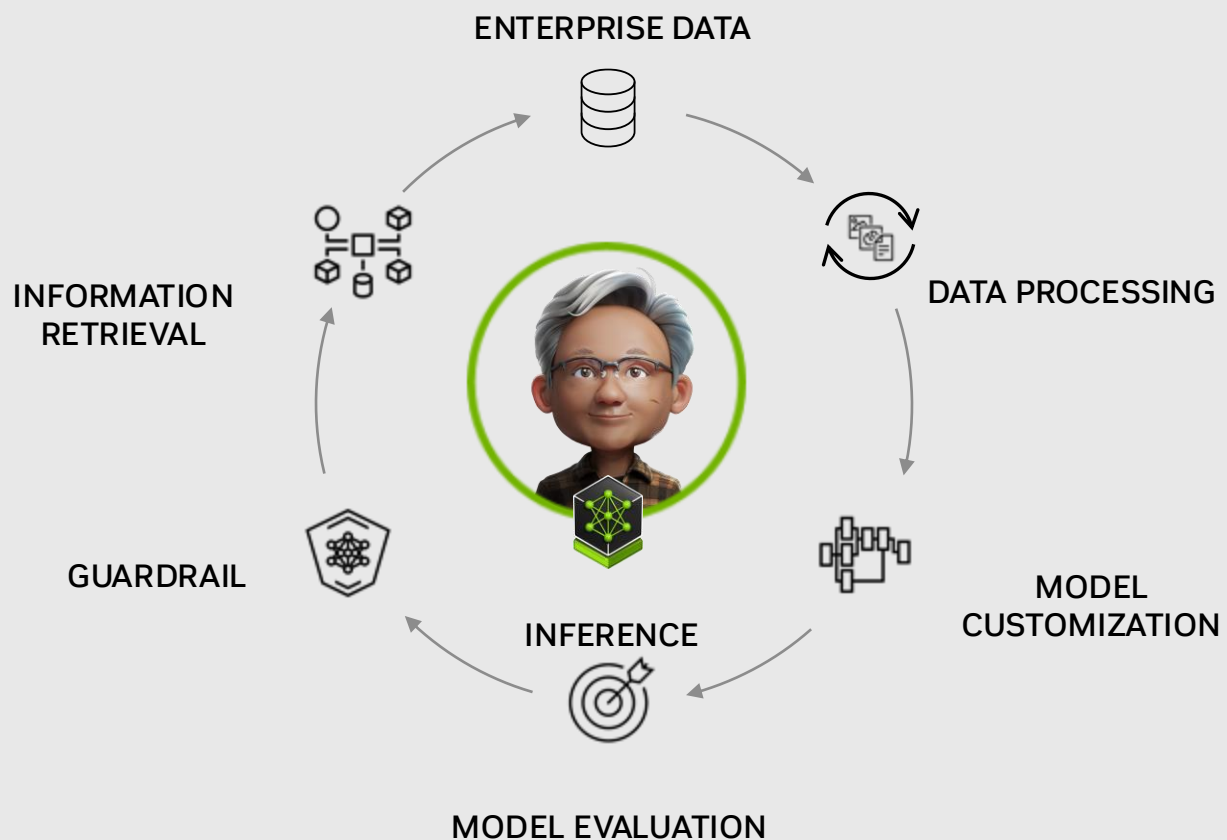
Rising Accuracy Targets

Continuous accuracy, performance optimizations at scale are hard and time consuming



Data Flywheels Unlock Agent Optimization

But existing tooling make them hard to implement



Complex Setup

End-to-end data flywheel deployments require managing 100s of tools and libraries



Discrete Software

Incompatible frameworks block seamless pipeline integration



Community Support

OSS lack prompt security resolutions and technical support



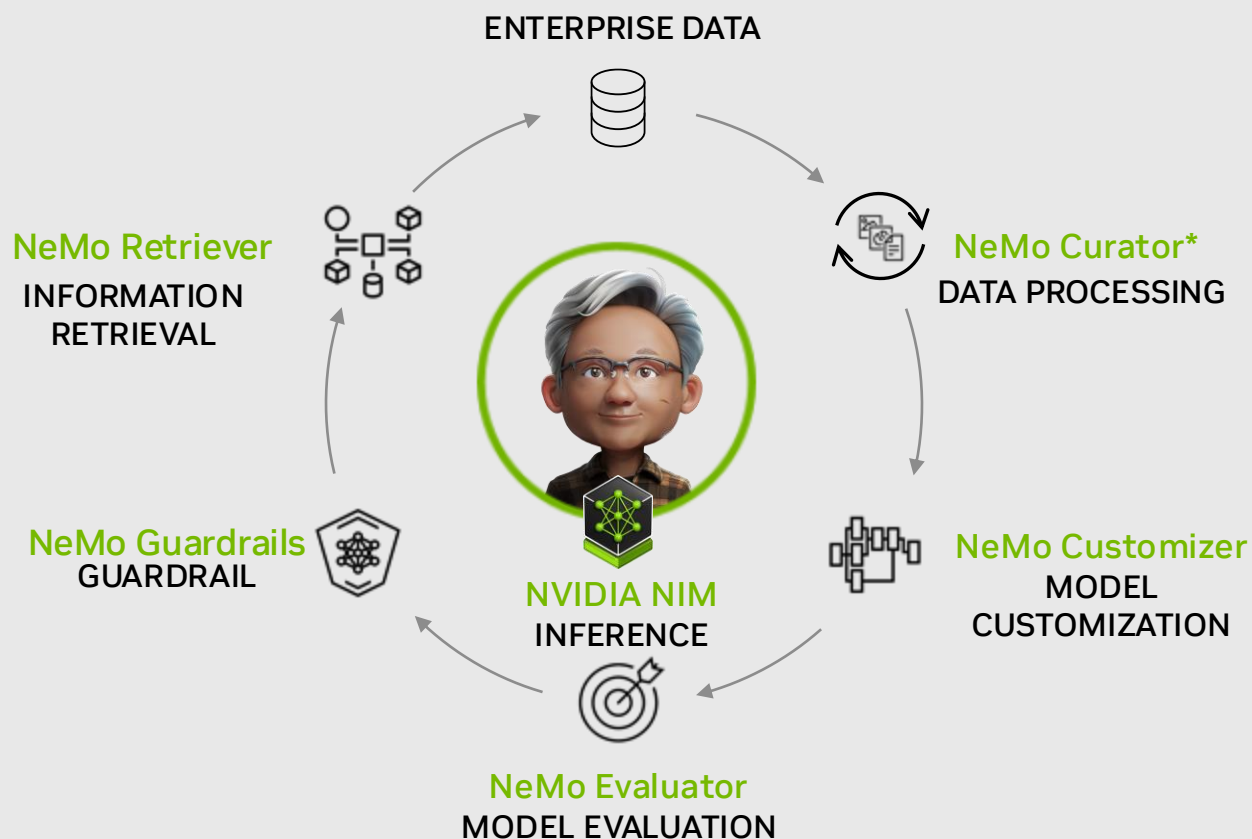
Vendor Lock-in

Enterprise solutions confine data flywheel to their ecosystem



NVIDIA NeMo Microservices

Modular End-to-End Platform to easily build Data Flywheels



Easy Setup

Modular microservices, deployable with standard APIs



Broad Ecosystem Support

Integrated in popular open frameworks and AI software platforms



Enterprise-grade

Secure, stable, and supported software



Run Anywhere

Provides higher security, privacy, and flexibility

NVIDIA NeMo Curator

Python SDK for scalable, configurable pipelines to curate text, image and video datasets for higher model accuracy



Efficient: Improve accuracy with less data and compute



Fast: Accelerated data processing



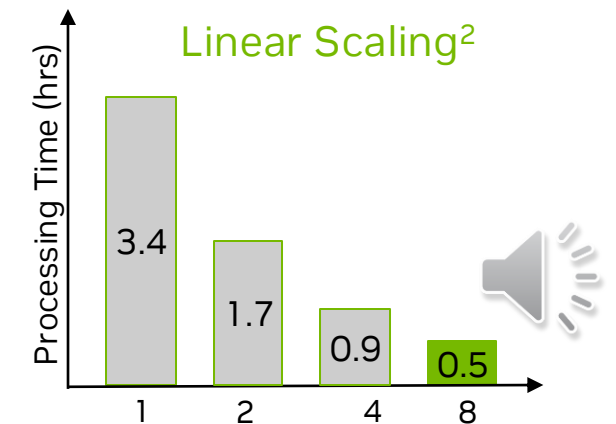
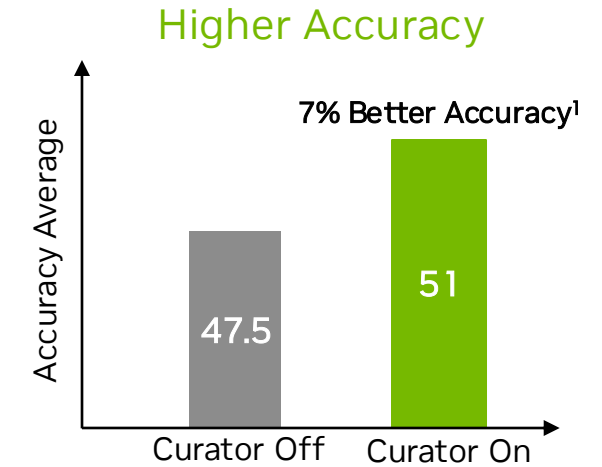
Scalable: Easily scale to process large datasets



Robust: SOTA classifier models for safety, content, and diversity



Secure: Run anywhere - Python APIs in a customizable and modular OSS library



Note:

1. 357M GPT model. Data curation methods: deduplication, heuristic filtering, and unicode correction. Accuracy: Average of Race, PiQA, Winogrande, and Hellaswag

2. Scaling Fuzzy Deduping 1.2T tokens RedPajama-v1 dataset. H100 80GB nodes

NVIDIA NeMo Customizer

Model customization microservice to integrate into AI platforms



Seamless customization: SoTA techniques such as LoRA, SFT (Full and Embedding Models), Knowledge Distillation, with DPO and GRPO upcoming.



Simple: 1 API call to model customization

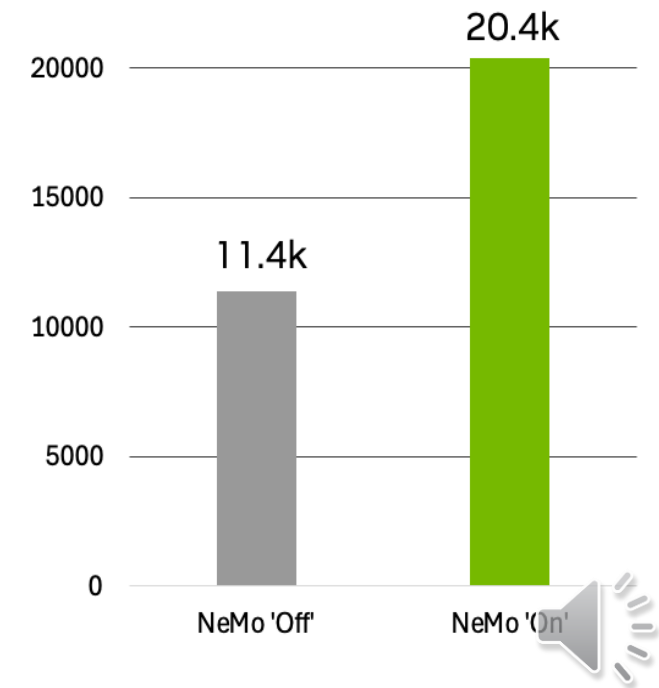


Wide coverage: Supports frontier model families including Llama, Mistral, Phi, Gemma, and Nemotron



Secure: Runs anywhere - on-prem and in the cloud with Kubernetes

1.8X Faster Post-Training



NVIDIA NeMo Evaluator

1 API call to evaluate 20+ academic benchmarks & custom metrics



Flexible: Full control of LLMs and AI pipelines evals for custom and standard benchmarks



Simple: Run evals at scale with a single API call anywhere with full data control and maintain CI/CD



Repeatable: Run consistent evals across teams with varying benchmark versioning using config files

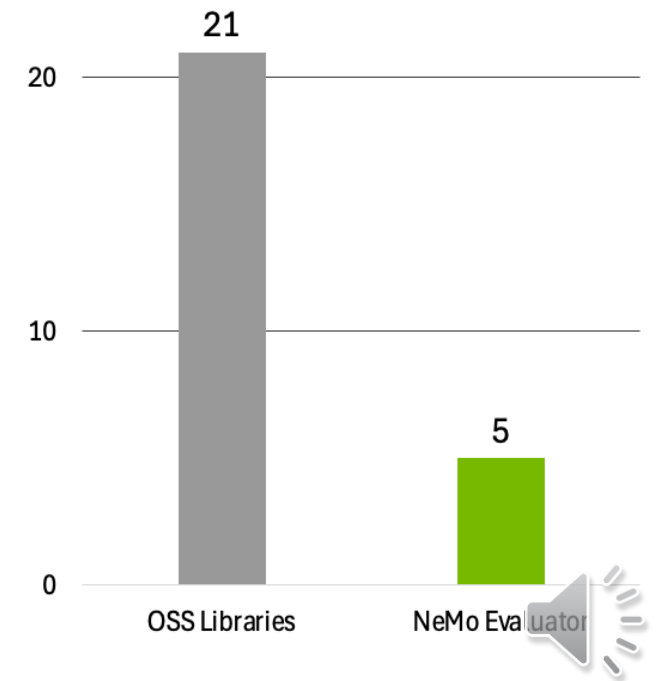


Top-of-tree: Enterprise-grade, supports latest benchmarks with security vulnerability patching



Secure: Runs anywhere - on-prem and in the cloud with Kubernetes

3X Reduction in APIs



Notes: OSS Libraries: Require 21 steps to do e2e evaluation of a customization job. NeMo : Only takes 5 API calls

NVIDIA NeMo Guardrails

Open-source, enterprise grade microservice to meet safety & security for your Gen AI solution



Industry Leading Guardrail: Content moderation, reduce hallucinations, toxicity, off-topic dialogs, PII-breach, and security jailbreaks



Customizable: Build rails targeted for your specific use-cases while adhering to own enterprise policies



Rail Orchestration: Rails helps to improve protection rate while reducing latency, increasing throughput and lowering costs

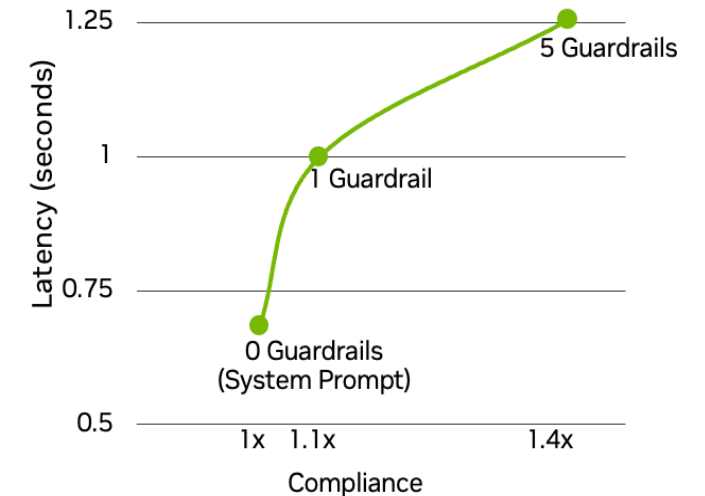


Easy Integration: Compatible with 3rd party APIs, such as ActiveFence, Fiddler, DataRobot etc.



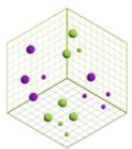
Native Integration: Popular Gen AI app developer tools integration (LangChain & LlamaIndex)

1.4X Higher Safety Compliance with Minimal Latency



NVIDIA NeMo Retriever

NVIDIA microservices power multimodal RAG pipelines turning data into knowledge



Lower Latency: Quickly parse, extract, and embed data from complex documents at scale



Customizable: State-of-the-art, customizable models, fine-tuned for accuracy

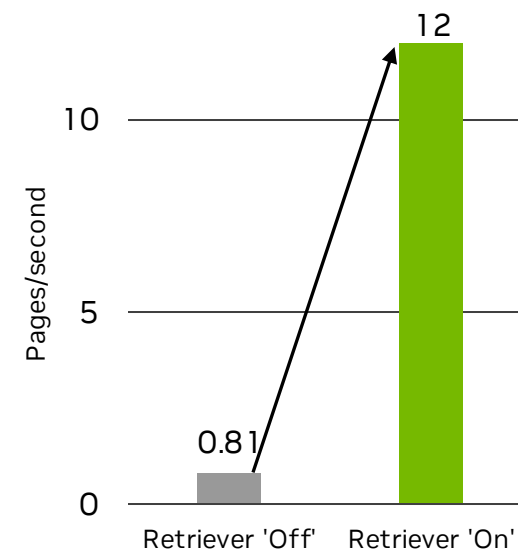


Enterprise-grade: Built on NVIDIA NIM Flexible, modular deployment and production ready



Secure: Runs anywhere, on-prem and in the cloud with Kubernetes and Slurm

15x Higher Throughput



Industry Pioneers Amp Digital Teammate Accuracy With NeMo Microservices



Boosts agent accuracy by up to 40%



Up to 40% fewer errors, 10x faster response





AI Agents Unlocked: CACEIS Redefines Client Conversations with VAST Data and NVIDIA

CACEIS works with VAST and NVIDIA NeMo Data Flywheel to help agents continuously improve

Broad Partner Ecosystem

Enterprises Can Use NeMo Microservices via a large ecosystem

MODELS

Google ∞ Meta Microsoft

MISTRAL
AI

NVIDIA

AI SOFTWARE & FRAMEWORKS

CLOUDERA

cremai

DATADOG

data
iku

DataRobot

DATASTACK

Haystack
by deepset

LangChain

LlamaIndex

∞ Meta

SuperAnnotate

Weights & Biases

ENTERPRISE PLATFORMS

amdocs

cādence

COHESITY

SAP

servicenow

SYNOPSYS

GLOBAL SYSTEM INTEGRATORS

accenture

Deloitte.

EY

HYBRID CLOUD

CISCO

DELL Technologies

Hewlett Packard
Enterprise

Lenovo

AI DATA PLATFORMS

ddn

DELL Technologies

Hewlett Packard
Enterprise

Hitachi Vantara

IBM

NetApp

NUTANIX

PURESTORAGE

VAST

WEKA

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Enterprise Blueprints

Customizable reference workflows with best practices for enterprise performance and scale



nvidia
Refine AI Agents through Contin...
Build a data flywheel, with NVIDIA NeMo microservices, that continuously optimizes AI...

blueprint data flywheel nim +4



nvidia
Build Digital Twins for AI Factory ...
Design, test, and optimize a new generation of intelligence manufacturing data centers usin...

ai factory blueprint +4



nvidia
Build an AI Agent for Enterprise ...
Build artificial general agents (AGA) powered by AGI models that continuously process and...

blueprint llama nemotron +7



nvidia
AI Weather Analytics with Earth-2
Develop AI powered weather analysis and forecasting application visualizing multi...

ai weather prediction blueprint +5



nvidia
Synthetic Manipulation Motion G...
Generate exponentially large amounts of synthetic motion trajectories for robot...

blueprint humanoids +10

Developer Examples

Interactive tutorials and example patterns to bootstrap AI development



nvidia
Bring LLMs to NIM
Use NIM to deploy a broad range of LLMs from Hugging Face.

blueprint nvidia ai nvidia



nvidia
Biomedical AI-Q Research Agent Blu...
Build advanced AI agents within the biomedical domain using the AI-Q Blueprint and the BioNeMo...

launchable agent blueprint +4



wandb
AI Observability for Data Flywheel
Streamline evaluation, monitoring, and optimization of AI data flywheel with Weights & Biases.

ai agents data flywheel wandb +5



iguazio
AI Orchestration for Data Flywheel
Orchestrate AI agents for data flywheel with MLRun and NVIDIA NeMo microservices.



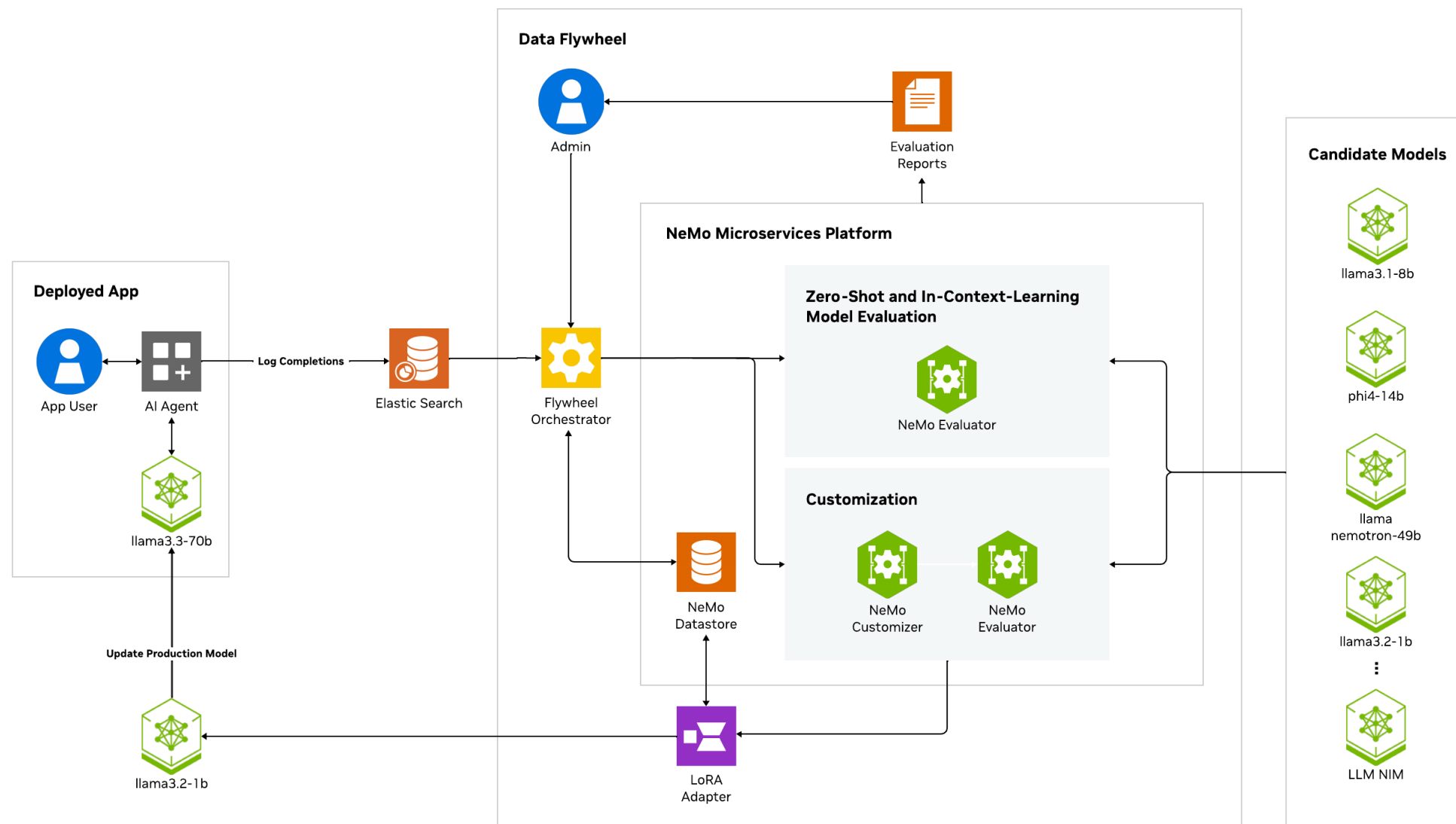
nvidia
Safety for Agentic AI
Improve safety, security, and privacy of AI systems at build, deploy and run stages.



nvidia
AI Agent for Telecom Network Confi...
Automate and optimize the configuration of radio access network (RAN) parameters using agentic AI...

NVIDIA AI Blueprint for Building Data Flywheels

Set-up Self-Optimizing Agentic Workflow With Ease Powered by NeMo Microservices



Blueprint Launchable

Pre-curated example demonstrating the data flywheel

Experience the Data Flywheel
to optimize a CX Agent for 'Tool-Calling'

- Deploy : Sample [Jupyter Notebook](#) via the launchable
- Goal : Find smaller, more efficient NIM that can match the tool-calling accuracy of the base llama 3.3-70b
- Candidate NIM :
 - llama 3.2-1b
 - llama 3.2-3b
 - llama 3.1-8b

The screenshot shows the NVIDIA Blueprint Launchable interface for a blueprint named 'data-flywheel'. At the top, there's a navigation bar with links to Explore, Models, Blueprints, GPUs, Docs, and Forum. The blueprint name 'data-flywheel' is displayed with a price tag of '\$12.07/hr' and a 'Deploy Launchable' button. Below this, a configuration bar shows the Compute section with 'NVIDIA A100 (80GiB)' and '2 GPUs x 24 CPUs | 340GiB', the Container section with 'VM Mode w/ Jupyter' and '- Preinstall Jupyter', the Files section with 'Notebook: data-flywheel-bp-tutorial' and a 'View Source' link, and the Exposed Ports section with 'Tunnels: mlflow:5000'. A 'Content Preview' section follows, stating 'This jupyter notebook will be present in the instance after deployment'. The main content area features a title 'Discover Cost-Efficient AI Customer Service Agents with NVIDIA Data Flywheel Blueprint', a 'Deploy Now' button, and a description: 'In this notebook, you will learn how to use the Data Flywheel Blueprint to continuously discover and promote more cost-efficient agents for an AI virtual customer service assistant.' Below this, there's a section for 'Data Flywheel Blueprint' with a small image, and another section for 'AI Virtual Assistant for Customer Service' with a description: 'The primary customer service agent in the AI Virtual Assistant uses tool calling to route user queries to specialized assistants, including:' followed by a list: '• Product Q&A' and '• Order status verification'.

Click to watch demo



Step 1: Load Sample Data

First, we need to import required libraries and configure pandas display options for better readability in notebook outputs.

```
[13]: import sys
from pathlib import Path
import requests
import time
from datetime import datetime
import json
import pandas as pd
from IPython.display import display, clear_output

pd.set_option('display.max_columns', None) # Show all columns
pd.set_option('display.width', None) # Width of the display in characters
pd.set_option('display.max_colwidth', None) # Show full content of each cell
```

Use the provided sample dataset from AI Virtual Assistant (aiva) (data/aiva_primary_assistant_dataset.jsonl) to simulate real user logs captured while an agentic customer service agent application is running. Each data point has the following schema:

Field	Type	Description
timestamp	int (epoch secs)	Time the request was issued
workload_id	str	Stable identifier for the logical task / route / agent node
client_id	str	Identifier of the application or deployment that generated traffic
request	dict	Exact <code>openai.ChatCompletion.create</code> payload received by the model
response	dict	Exact <code>ChatCompletion</code> response returned by the model

The `request` uses the OpenAI `ChatCompletions` request format and contains the following attributes:

- `model`: includes the Model ID used to generate the response.
- `messages`: includes a `system` message as well as a `user` query.
- `tools`: includes a list of functions and parameters available to the LLM to choose from, as well as their parameters and descriptions.

```
[14]: DATA_PATH = data_dir / "aiva_primary_assistant_dataset.jsonl"
```

```
!head -n1 {DATA_PATH} | jq
```

```
{
  "function": {
    "name": "HandleOtherTalk",
    "description": "Handles greetings and other absurd queries by offering polite redirection and clearly explaining the limitations of the chatbot.",
    "parameters": {
```

Get Started Now

Continuously Optimize your Models NVIDIA AI Data Flywheel Blueprint

The screenshot shows the NVIDIA AI Data Flywheel Blueprint deployment interface. At the top, there's a navigation bar with links: NVIDIA, Explore, Models, Blueprints, GPUs, Docs, and Forum. The main heading is "data-flywheel" with a price tag of "\$12.07/hr" and a "Deploy Launchable" button. Below this, a configuration bar shows: Compute (NVIDIA A100 80GiB, 2 GPUs x 24 CPUs | 340GiB, 512GiB, GCP), Container (VM Mode w/ Jupyter, Preinstall Jupyter), Files (Notebook: data-flywheel-bp-tutorial, View Source), and Exposed Ports (Tunnels: mlflow:5000). A "Content Preview" section follows, stating "This jupyter notebook will be present in the instance after deployment". The preview content is titled "Discover Cost-Efficient AI Customer Service Agents with NVIDIA Data Flywheel Blueprint" and includes a "Deploy Now" button. The text below the button says: "In this notebook, you will learn how to use the Data Flywheel Blueprint to continuously discover and promote more cost-efficient agents for an AI virtual customer service assistant."

Deploy the Blueprint: build.nvidia.com/blueprints

GitHub: [NVIDIA-AI-Blueprints/data-flywheel](https://github.com/NVIDIA-AI-Blueprints/data-flywheel)

